

1. Theoretical developments

Let $(M, \mathcal{O}, \mathcal{A})$ be a smooth manifold.

- (a) Show that the set $C^\infty(M) := \{f : M \rightarrow \mathbb{R} \text{ smooth}\}$ equipped with a pointwise defined addition $\boxplus$ and S-multiplication $\boxtimes$ constitutes a vector space $(C^\infty(M), \boxplus, \boxtimes)$. Can you find an infinite set E of elements in $C^\infty(M)$ such that every finite subset $F \subset E$ is linearly independent? What does this imply for the dimension of the vector space?

Solution: The pointwise addition of two functions $f, g \in C^\infty(M)$ is defined through

$$(f \boxplus g)(p) := f(p) + g(p)$$

and the pointwise S-multiplication through

$$(\lambda \boxtimes f)(p) := \lambda f(p)$$

for all $p \in M$, where where the addition and multiplication on the respective right hand sides are those in $\mathbb{R}$. Identifying $O_{C^\infty(M)}$ as the function defined by $O(p) := 0$ and the additive inverses $-f$ as the functions defined by $(-f)(p) := -f(p)$ for all $p \in M$, all eight vector space axioms directly follow from the nine axioms satisfied by the field $(\mathbb{R}, +, \cdot)$ of real numbers. Specializing, for simplicity, to $(M, \mathcal{O}) = (\mathbb{R}, \mathcal{O}_s)$ and choosing (M, id_M) as the only chart that defines an atlas $\mathcal{A}$, one has the smooth manifold $(M, \mathcal{O}_s, \mathcal{A})$. The smooth functions on M are then identical to the smooth functions on $\mathbb{R}$ and one can choose an infinite number of linearly independent smooth functions, e.g.,

$$f_n(x) := x^n.$$

Thus $C^\infty(\mathbb{R})$ is infinite-dimensional. Extension of the argument to any smooth manifold $(M, \mathcal{O}, \mathcal{A})$ is most elegantly done using a partition of unity, which is beyond the scope of this course (sorry) but an easy concept to look up.

- (b) For a smooth manifold $(M, \mathcal{O}, \mathcal{A})$, the set $T_p M := \{X_{\gamma,p} | \gamma : \mathbb{R} \rightarrow M \text{ with } \gamma(0) = p\}$ of tangent vectors at a point $p \in M$ can be equipped with a map $\boxplus : T_p M \times T_p M \rightarrow T_p M$ given by letting

$$X_{\gamma,p} \boxplus X_{\delta,p} := X_{\alpha,p},$$

where the curve α on M is defined in terms of the curves γ and δ on M as

$$\alpha(\lambda) = x^{-1}((x \circ \gamma)(\lambda) + (x \circ \delta)(\lambda) - x(p))$$

with respect to any chart $(U, x) \in \mathcal{A}$ with $p \in U$. Prove that choosing any other chart $(V, y) \in \mathcal{A}$ with $p \in V$ yields the same tangent vector $X_{\gamma,p}$ and thus the same map $\boxplus$. Why is this important?

Solution: It is easy to see that $\alpha(0) = x^{-1}((x \circ \gamma)(0) + (x \circ \delta)(0) - x(p)) = x^{-1}(x(p)) = p$. Thus for any $f \in C^\infty(M)$, one has

$$X_{\alpha,p} f = (f \circ \alpha)'(0) = ((f \circ x^{-1}) \circ (x \circ \alpha))'(0) = \sum_{a=1}^{\dim M} (x \circ \alpha)'(0) \cdot \partial_a (f \circ x^{-1})((x \circ \alpha)(0)).$$

Now since

$$(x \circ \alpha)(0) = x(p) \quad \text{and} \quad (x \circ \alpha)'(0) = (x \circ \gamma)'(0) + (x \circ \delta)'(0)$$

one has

$$X_{\alpha,p}f = \sum_{a=1}^{\dim M} ((x \circ \gamma)'(0) + (x \circ \delta)'(0)) \left(\frac{\partial f}{\partial x^a} \right)_p = X_{\gamma,p} \diamond X_{\delta,p}$$

without any constraint on the choice of the employed chart (U, x) other than that $p \in U$. Thus exactly the same calculation holds for any other chart (V, y) with $p \in V$. This is important because a definition – like the one of $\diamond$ here – that uses “some” chart must ultimately be independent of the particularly chosen chart in order to have a meaning, since the choice of charts is largely arbitrary.

- (c) Let X be an element of the tangent bundle TM . Explain why choosing a chart $(U, x) \in \mathcal{A}$ with $\pi(X) \in U$ and another chart (V, y) with $\pi(X) \in V$ allows to write, alternatively,

$$X = \sum_{a=1}^{\dim M} X_{(U,x)}^a \left(\frac{\partial}{\partial x^a} \right)_{\pi(X)} \quad \text{or} \quad X = \sum_{a=1}^{\dim M} X_{(V,y)}^a \left(\frac{\partial}{\partial y^a} \right)_{\pi(X)}$$

for unique real numbers $X_{(U,x)}^1, \dots, X_{(U,x)}^{\dim M}$ and $X_{(V,y)}^1, \dots, X_{(V,y)}^{\dim M}$, respectively. Show from these two alternative expansions of X in terms of the two chart-induced bases that the respective components are related as

$$X_{(U,x)}^a = \sum_{b=1}^{\dim M} \left(\frac{\partial x^a}{\partial y^b} \right)_{\pi(X)} X_{(V,y)}^b.$$

Solution: Up to the last expression, the set-up of this entire problem is a pure composition of definitions that allows to frame the question in terms of an element X of the tangent bundle rather than immediately focusing on a particular tangent space $T_p M$ to which, in the end, everything boils down. Also an important exercise to cut through the thicket to see the essence of a problem. Mean, but important. The central technical point that remains is to show that

$$\left(\frac{\partial f}{\partial y^a} \right)_{\pi(X)} = \left(\frac{\partial x^b}{\partial y^a} \right)_{\pi(X)} \left(\frac{\partial f}{\partial x^b} \right)_{\pi(X)}$$

for any $f \in C^\infty(M)$. But this is simply a matter of using the definition of these symbols and the multi-dimensional chain rule in order to rewrite the left hand side of this above relation as

$$\left(\frac{\partial f}{\partial y^a} \right)_{\pi(X)} = \partial_a(f \circ y^{-1})(y(p)) = \partial_a((f \circ x^{-1}) \circ (x \circ y^{-1}))(y(p)) = \partial_a(x^b \circ y^{-1})(y(p)) \partial_b(f \circ x^{-1})(x(p)),$$

for $p = \pi(X)$. But the right hand side of this string of equations is already the right hand side of the equality that was to be shown.

- (d) Let $\mathcal{X} : M \rightarrow TM$ and $\mathcal{Y} : M \rightarrow TM$ be two vector fields on M and $f \in C^\infty(M)$. First show that

$$\mathcal{X}f : M \rightarrow \mathbb{R}, \quad (\mathcal{X}f)(p) := \mathcal{X}(p)f$$

is a smooth function on M . Now consider the so-called *Lie bracket* $[\mathcal{X}, \mathcal{Y}]$ defined by its action of an arbitrary function $f \in C^\infty(M)$ through

$$[\mathcal{X}, \mathcal{Y}]f := \mathcal{X}(\mathcal{Y}f) - \mathcal{Y}(\mathcal{X}f).$$

Obviously, $[\mathcal{X}, \mathcal{Y}]f$ is a smooth function on M since both $\mathcal{X}(\mathcal{Y}f)$ and $\mathcal{Y}(\mathcal{X}f)$ are. However, show that while neither $\mathcal{X}\mathcal{Y}$ nor $\mathcal{Y}\mathcal{X}$ are vector fields, the Lie bracket $[\mathcal{X}, \mathcal{Y}]$ is a vector field on M (Hint: Evaluate $\mathcal{X}(\mathcal{Y})f$ and $\mathcal{Y}(\mathcal{X})f$ at some point p in the domain U of some chart $(U, x) \in \mathcal{A}$ and explain why only their difference, but none of them individually, presents a directional derivative of f at p).

Solution: Using the suggested chart, one has for any $f \in C^\infty(M)$ that

$$\mathcal{Y}f = \mathcal{Y}_{(U,x)}^a \left(\frac{\partial f}{\partial x^a} \right)$$

for component functions $\mathcal{Y}_{(U,x)}^a : U \rightarrow \mathbb{R}$ and thus

$$\mathcal{X}(\mathcal{Y}f) = \mathcal{X}_{(U,x)}^b \left(\frac{\partial}{\partial x^b} \right) \left[\mathcal{Y}_{(U,x)}^a \left(\frac{\partial f}{\partial x^a} \right) \right] = \mathcal{X}_{(U,x)}^b \left(\frac{\partial \mathcal{Y}_{(U,x)}^a}{\partial x^b} \right) \left(\frac{\partial f}{\partial x^a} \right) + \mathcal{X}_{(U,x)}^b \mathcal{Y}_{(U,x)}^a \left(\frac{\partial}{\partial x^b} \right) \left(\frac{\partial f}{\partial x^a} \right),$$

and analogously

$$\mathcal{Y}(\mathcal{X}f) = \dots = \mathcal{Y}_{(U,x)}^b \left(\frac{\partial \mathcal{X}_{(U,x)}^a}{\partial x^b} \right) \left(\frac{\partial f}{\partial x^a} \right) + \mathcal{Y}_{(U,x)}^b \mathcal{X}_{(U,x)}^a \left(\frac{\partial}{\partial x^b} \right) \left(\frac{\partial f}{\partial x^a} \right).$$

It is clear from the respective right hand sides of these two relations that the left hand side cannot be understood as the action of some vector field on the function f , since only the respective first terms on the right hand side are directional derivatives of the function f , while the respective last terms contain some kind of second derivative of the function. It is easy to see that

$$\left(\frac{\partial}{\partial x^b} \right) \left(\frac{\partial f}{\partial x^a} \right) = \partial_b (\partial_a (f \circ x^{-1}) \circ x \circ x^{-1}) = \partial_b \partial_a (f \circ x^{-1}) = \partial_a \partial_b (f \circ x^{-1}) = \left(\frac{\partial}{\partial x^a} \right) \left(\frac{\partial f}{\partial x^b} \right);$$

Thus the difference

$$[\mathcal{X}, \mathcal{Y}]f = \left(\mathcal{X}_{(U,x)}^b \left(\frac{\partial \mathcal{Y}_{(U,x)}^a}{\partial x^b} \right) - \mathcal{Y}_{(U,x)}^a \left(\frac{\partial \mathcal{X}_{(U,x)}^b}{\partial x^b} \right) \right) \left(\frac{\partial f}{\partial x^a} \right)$$

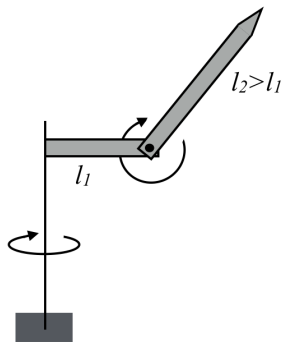
acts on f like a directional derivative at each point $p \in U$ so that

$$[\mathcal{X}, \mathcal{Y}]^a = \mathcal{X}_{(U,x)}^b \left(\frac{\partial \mathcal{Y}_{(U,x)}^a}{\partial x^b} \right) - \mathcal{Y}_{(U,x)}^a \left(\frac{\partial \mathcal{X}_{(U,x)}^b}{\partial x^b} \right)$$

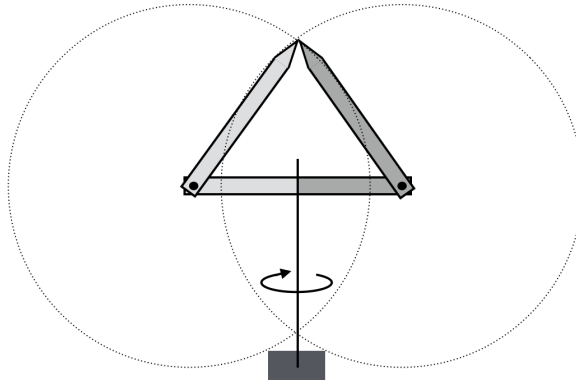
are the component functions of the vector field $[\mathcal{X}, \mathcal{Y}]$.

2. Practical application: A simple robot whose description *requires* a manifold

Consider a robot with a short arm of length l_1 mounted on a vertical axis and a longer arm of length $l_2 > l_1$ mounted on a perpendicular axis at the end of the short arm, see figure (a).



(a)



(b)

- (a) Choosing the mounting point of the short arm on the vertical axis as the origin, define the set $P \subset \mathbb{R}^3$ of all possible points that can be reached by the tip of the long arm, if the distance of any two points in $\mathbb{R}^3$ is given by

$$\|(a, b, c) - (d, e, f)\| := \sqrt{(a-d)^2 + (b-e)^2 + (c-f)^2}.$$

Solution:

$$P = \{(a_1 + a_2 \frac{a_1}{l_1}, b_1 + a_2 \frac{b_1}{l_1}, b_2) \in \mathbb{R}^3 \mid a_1^2 + b_1^2 = l_1^2 \text{ and } a_2^2 + b_2^2 = l_2^2\}.$$

- (b) Considering figure (b), explain why P fails to represent all possible the configurations of the robot.

Solution: *There are two points in P , namely*

$$(0, 0, \pm\sqrt{l_2^2 - l_1^2}),$$

to each of which corresponds an uncountable infinity of different configurations.

- (c) Define the set $C_1 := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = l_1^2\}$ and consider it equipped with the subset topology $\mathcal{O}_{|\mathbb{R}^2}$ defined in problem 2(a) of training set 3. Provide two charts (U, x) and (V, y) that cover C_1 (which simply means that $C_1 = U \cup V$) and reveal C_1 as a one-dimensional topological manifold.

Solution: *Choose $U := C_1 \setminus \{0, l_1\}$ and $V := C_1 \setminus \{0, -l_1\}$ with the respective chart maps defined by*

$$x(\alpha, \beta) := \frac{\alpha}{l_1 - \beta} \quad \text{and} \quad y(\alpha, \beta) = \frac{\alpha}{l_1 + \beta}.$$

These are continuous maps and indeed $U \cup V = C_1$.

- (d) Check whether the two charts you constructed in (b) constitute a smooth atlas. If not, replace one of the two charts by another one such that the atlas becomes smooth.

Solution: *One needs to check the smoothness of $x \circ y^{-1}$ and $y \circ x^{-1}$. The respective inverse maps are easily verified to be*

$$(\alpha(x), \beta(x)) = (l_1 \frac{2x}{1+x^2}, l_1 \frac{x^2-1}{1+x^2}) \quad \text{and} \quad (\alpha(y), \beta(y)) = (l_1 \frac{2y}{1+y^2}, l_1 \frac{1-y^2}{1+y^2}).$$

Thus

$$(x \circ y^{-1})(v) = \frac{\frac{2v}{1+v^2}}{1 - \frac{1-v^2}{1+v^2}} = \frac{1}{v} \quad \text{and} \quad (y \circ x^{-1})(u) = \frac{\frac{2u}{1+u^2}}{1 + \frac{u^2-1}{1+u^2}} = \frac{1}{u}$$

for all $v \in y(U \cap V) = \mathbb{R} \setminus \{0\}$ and $u \in x(U \cap V) = \mathbb{R} \setminus \{0\}$. Since these latter two maps are smooth on their domains, the charts from (c) constitute a smooth atlas.

- (e) Now consider the set $C_1 \times C_2$, where C_1 is as above and $C_2 := \{(z, w) \in \mathbb{R}^2 \mid z^2 + w^2 = l_2^2\}$. Consider the set $C_1 \times C_2$ equipped with the product topology $\mathcal{O}_{C_1 \times C_2}$ defined in problem 2(b) of training set 3. Use the two charts for C_1 from part (c) and two analogously constructed charts for C_2 in order to construct *four* charts that constitute a smooth atlas $\mathcal{A}_{C_1 \times C_2}$ for the topological manifold $(C_1 \times C_2, \mathcal{O}_{C_1 \times C_2})$.

Solution: *Introduce charts (U', x') and (V', y') on C_2 in complete analogy to the charts (U, x) and (V, y) on C_1 . Then let $U \times U'$, $U \times V'$, $V \times U'$, $V \times V'$ be four chart domains, which obviously cover $C_1 \times C_2$ and define the chart map*

$$x_{U \times U'} : U \times U' \rightarrow \mathbb{R}^2, \quad (p, q) \mapsto (x(p), x'(q))$$

on the domain $U \times U'$ and analogous chart maps on the three other domains.

These are continuous maps because of the choice of the product topology on $C_1 \times C_2$. The various transition functions between those four charts are smooth, as can be deduced from the smoothness of the various transition functions on C_1 and C_2 shown in (d).

- (f) What is the dimension of the thus defined smooth manifold $(C_1 \times C_2, \mathcal{O}_{C_1 \times C_2}, \mathcal{A}_{C_1 \times C_2})$? Explain why $C_1 \times C_2$ represents the configuration space of the robot and how it differs from the set P defined above.

Solution: Since the charts all map into regions of $\mathbb{R}^2$, the manifold is two-dimensional and represents all configurations since its points correspond to all possible combined positions of the two arms, in contrast to the set P . This manifold $C_1 \times C_2$, which is the smooth product manifold of two circles, is known as the 2-torus.

- (g) Provide a smooth curve γ in $C_1 \times C_2$ that connects the two different configurations of the robot shown in figure (b) such that the tip of the long arm never moves. Identify all charts through whose domains this curve runs and provide the respective chart representations of the curve.

Solution: The configurations shown in figure (b) are those pointing at the point $(0, 0, \sqrt{l_2^2 - l_1^2}) \in \mathbb{R}^3$ and correspond to the two points

$$p_{\pm} := (\pm l_1, 0, l_1, \sqrt{l_2^2 - l_1^2}) \in C_1 \times C_2$$

and the curve

$$\gamma : [0, \pi] \rightarrow C_1 \times C_2, \quad \phi \mapsto (l_1 \cos \phi, l_1 \sin \phi, 0, l_1, \sqrt{l_2^2 - l_1^2})$$

connects these two points. The entire curve lies, for instance, in the chart domain $V \times U'$ (or, alternatively, in $V \times V'$) defined in parts (d) and (e), with the chart map

$$x_{V \times U'} : V \times U' \rightarrow \mathbb{R}^2, \quad (p, q) \mapsto (x(p), y'(q)).$$

Thus the representative of the curve γ in this chart is the map

$$x_{V \times U'} \circ \gamma : [0, \pi] \rightarrow \mathbb{R}^2, \quad \phi \mapsto \left(\frac{l_1 \cos \phi}{l_1 + l_1 \sin \phi}, \frac{l_1}{l_2 - \sqrt{l_2^2 - l_1^2}} \right).$$

No further chart is needed.

- (h) Calculate the tangent vectors to the curve γ at the two points corresponding to the two configurations shown in figure (b).

Solution: Those two points correspond to the parameter values $\phi = 0$ and $\phi = \pi$. With respect to the chart chosen in part (g), the components of the tangent vector at $\gamma(0)$ and $\gamma(\pi)$ are

$$(x_{V \times U'} \circ \gamma)'(0) = (-1, 0) \quad \text{and} \quad (x_{V \times U'} \circ \gamma)'(\pi) = (1, 0).$$

Thus the actual tangent vectors are

$$X_{\gamma, \gamma(0)} = (-1) \cdot \left(\frac{\partial}{\partial y} \right)_{\gamma(0)} + 0 \cdot \left(\frac{\partial}{\partial x'} \right)_{\gamma(0)} = - \left(\frac{\partial}{\partial y} \right)_{\gamma(0)} \quad \text{and} \quad X_{\gamma, \gamma(\pi)} = \left(\frac{\partial}{\partial y} \right)_{\gamma(\pi)}.$$

- (i) Do you see how badly needed smooth manifolds are in order to describe the configuration space (and much more!) for even the simplest robots?

Solution: Yes. Wow. Geez. So true.